MUHAMMAD HANNAN HARIS

Cloud Engineer | AWS Certified Solutions Architect

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PROFESSIONAL SUMMARY

Senior Cloud & AI Engineer with 2.5 years of hands-on experience designing, provisioning, and maintaining secure cloud infrastructure on AWS, with practical exposure to Azure and Alibaba Cloud environments. AWS Certified Solutions Architect proficient in Infrastructure as Code (Terraform, Ansible), automated CI/CD workflows, containerized deployments, AI workflow automation (n8n, Milvus, RAG pipelines), and LLM dataset engineering. Demonstrated track record in reducing cloud expenditure, implementing resilient disaster recovery solutions, and delivering scalable cloud and AI systems across multinational teams.

TECHNICAL SKILLS

- **Cloud Platforms:** AWS (EC2, RDS, S3, VPC, CloudWatch, CloudTrail, Secrets Manager, DRS, Lightsail, Aurora, Lambda, ELB, Auto Scaling, WAF, GuardDuty), Azure (Application Gateway, WAF), Alibaba Cloud (ECS, OSS, VPC, RDS).
- **Infrastructure as Code & CI/CD:** Terraform, Ansible, Git, GitHub Actions, Jenkins, Automated Build/Test Pipelines.
- **Containers, Orchestration & Automation:** Docker, Docker Compose, Kubernetes basics, Dokploy, n8n.
- **Networking & Security:** VPC Design, OpenVPN, Security Groups, NACLs, Route 53, SSL/TLS, AWS WAF, IAM.
- **Databases & Storage:** MySQL, PostgreSQL, AWS RDS, Aurora, S3, EBS, EFS, AWS DMS.
- **Operating Systems & Scripting:** Linux (Ubuntu), Bash, Python, YAML, JSON, HCL, PowerShell.

PROFESSIONAL EXPERIENCE

Senior Cloud & AI Engineer May 2024 – Present

Puffersoft (Pvt.) Limited Faisalabad, Pakistan

- Designed and deployed a secure three-tier AWS architecture consisting of public and private subnets across multiple availability zones, Application Load Balancers, NAT gateways, and layered security groups and NACLs, supporting around 150,000 daily user requests.
- Built a scalable file upload and download system on S3 using multipart uploads and presigned URLs behind CloudFront, with CORS configured for the web client, handling files up to 5 GB and supporting approximately 2,000 daily uploads.
- Automated the detection and blocking of suspicious traffic using AWS WAF and GuardDuty, paired with EventBridge and Lambda for remediation, cutting manual incident response time from 45 minutes to near-real-time.
- Constructed automated CI/CD pipelines in GitHub Actions to run test suites and code linting, build Docker container images, and block deployments upon test failure, ensuring continuous delivery and deployment reliability.
- Deployed Alibaba Cloud ECS compute instances configured with Docker Compose under strict security rules, hosting self-hosted tools including n8n, PostgreSQL, pgAdmin, pgvector, Open WebUI, and NocoDB.
- Designed and automated the deployment of AIXWorkspace.net across AWS and Alibaba Cloud,

bringing together automation, databases, RAG, data management, and AI interfaces.
- Optimized monthly AWS compute spend by 22% through the strategic implementation of Reserved Instances, Spot Instances, and dynamic Auto Scaling policies across EC2 and RDS workloads.
- Engineered Infrastructure as Code modules using Terraform to automate network, compute, and security resource provisioning across environments.
- Executed migration of a 400 GB production MySQL database to AWS RDS utilizing AWS Database Migration Service (DMS) with minimal operational downtime.
- Integrated AWS Secrets Manager across microservices to manage application credentials securely and satisfy compliance standards.
- Developed an open-source custom Milvus node integration for n8n (github.com/Hannan-Haris/n8n-nodes-milvus-db), connecting workflow automation pipelines with vector database operations.
- Developed production-grade AI automation workflows using n8n, Qwen LLMs, and Milvus, delivering bilingual document approval (DocSign), multi-tier HR, Finance, and Executive approval routing, RAG-powered customer support, and human-reviewed content generation.
- Mentored 3 junior cloud engineers in AWS architectural best practices, Terraform infrastructure code development, and system troubleshooting techniques.

Data Scientist (Part-Time, Remote) June 2024 – January 2026
A.R Data Science Faisalabad, Pakistan (Remote)

- Built and cleaned high-quality datasets for large language models to support model training, testing, and overall performance optimization.
- Designed agentic reasoning tasks to improve model decision-making accuracy and multi-step problem-solving capabilities.
- Reviewed and validated AI-generated outputs to maintain strict data quality standards aligned with project objectives.
- Applied machine learning, statistical methods, and NLP techniques for exploratory data analysis, feature engineering, and dataset tuning to uncover trends and gaps.
- Constructed automated data pipelines to accelerate preprocessing workflows and integrated data-driven components into production AI systems.
- Supervised task pipelines and mentored junior data scientists while authoring clear technical reports and documentation for local and international teams.

KEY PROJECTS

AWS EBS IOPS Auto-Scaling Terraform, AWS Lambda, CloudWatch, EventBridge

- Developed a multi-region Terraform project that automatically increases EBS volume IOPS and throughput when CloudWatch performance alarms trigger, using EventBridge-driven Lambda functions and SNS notifications.
- Packaged the solution into reusable IAM, SNS, and Lambda infrastructure modules configured with S3 remote state and DynamoDB state locking.

Secure Remote Access Gateway AWS VPC, OpenVPN, Terraform

- Authored Infrastructure as Code templates to deploy OpenVPN gateways on AWS with strict security groups, private VPC routing, and MFA controls.

DeepFake Audio Detection System (Final Year Project) Python, Machine Learning, Audio Signal Processing

- Developed a machine learning-based audio classification system to detect synthetic and manipulated voice recordings by extracting acoustic features and evaluating model accuracy across spoofed audio datasets.

EDUCATION

Bachelor of Science in Computer Science May 2024
University of Engineering and Technology Lahore, Pakistan

Higher Secondary School Certificate (Pre-Engineering) 2020
BISE Faisalabad Faisalabad, Pakistan

CERTIFICATIONS

- **AWS Certified Solutions Architect – Associate**
- Relevant Training: Terraform Infrastructure Automation, Ansible Automation, Docker Fundamentals, Kubernetes Basics, n8n Automation.

SPEAKING & SEMINARS

Seminar Presenter – Building Your Career in the Age of AI November 2026
FAST-NUCES University Faisalabad, Pakistan

- Represented Puffersoft (Pvt.) Limited to deliver a technical and career guidance session on emerging AI technologies, industry expectations, and practical pathways for engineering students entering the modern tech landscape.

LANGUAGES

- **English:** Professional / IELTs: 6.5
- **Urdu:** Native